

BROWN FIELD, CA, USA

WMO: 722904

Lat: 32.572N

Lon: 116.979W

Elev: 157

StdP: 99.45

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 03178

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.9	5.4	-13.4	1.2	16.5	-10.5	1.6	16.8	9.1	16.2	8.2	15.8	2.7	120	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	32.2	17.9	29.9	18.1	28.0	18.2	21.9	28.0	21.2	26.6	20.5	25.4	3.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.0	24.0	19.2	14.2	23.4	18.7	13.8	23.0	64.8	28.1	62.1	26.6	59.7	25.6	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	5.8	5.3	DB	1.4	36.8	1.3	2.0	0.5	38.2	-0.3	39.4	-1.0	40.5	-2.0	41.9	(4)
(5)			WB	-1.3	23.1	1.4	1.0	-2.3	23.8	-3.1	24.4	-3.9	25.0	-4.9	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	13.4	13.1	14.3	15.4	17.1	18.9	21.1	21.7	21.4	19.1	16.0	13.1
	DBStd	4.08	3.10	2.88	2.60	2.61	2.21	1.92	2.02	2.01	2.67	2.83	3.13	2.72
	HDD10.0	13	4	3	1	0	0	0	0	0	0	0	1	4
	HDD18.3	865	155	149	131	94	51	13	1	1	2	23	84	162
	CDD10.0	2594	111	90	133	163	221	266	344	362	341	281	182	101
	CDD18.3	406	3	3	4	7	13	29	86	104	94	45	16	1
	CDH23.3	2572	55	49	76	114	122	116	367	508	571	387	183	24
CDH26.7	752	8	11	23	38	47	29	74	120	188	151	62	2	
Wind	WSAvg	2.3	2.4	2.7	2.4	2.6	2.5	2.3	2.1	2.0	2.0	2.1	2.3	2.5
Precipitation	PrecAvg	306	49	64	42	25	8	1	3	2	6	16	31	52
	PrecMax	568	127	256	94	50	40	5	22	5	31	142	89	160
	PrecMin	162	6	3	3	4	0	0	0	0	0	0	10	2
	PrecStd	109	40	58	24	15	11	1	5	2	8	32	16	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.8	28.2	29.6	31.1	32.2	30.5	31.2	32.6	35.0	34.3	31.7	26.2
		MCWB	13.0	13.4	14.3	14.0	15.2	19.4	21.3	20.7	19.9	16.5	14.2	12.0
	2%	DB	24.7	24.4	25.1	26.7	26.7	26.1	28.6	29.6	31.2	30.9	28.3	23.1
		MCWB	12.1	12.5	13.4	14.3	16.5	19.2	20.9	20.8	20.4	15.7	13.7	11.3
	5%	DB	22.3	21.3	22.1	23.0	23.0	24.0	26.9	27.7	28.6	27.6	24.9	21.1
		MCWB	11.5	11.9	13.5	14.2	16.3	18.2	20.2	20.6	20.1	15.7	12.8	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	19.6	18.6	19.4	20.8	21.4	22.7	25.2	26.2	26.6	24.7	22.2	18.7
		MCWB	11.1	11.7	13.4	14.4	15.8	17.5	19.6	20.1	19.6	15.9	12.5	11.3
	0.4%	WB	15.7	15.8	16.8	17.3	19.1	20.7	22.7	22.7	23.0	20.8	17.4	16.2
		MCDB	20.2	19.8	23.5	24.1	25.5	27.9	28.2	29.5	30.4	25.2	22.6	19.6
	2%	WB	14.7	14.6	15.6	16.3	17.9	19.5	21.5	21.7	22.0	19.3	16.3	14.9
		MCDB	18.8	19.3	21.0	22.5	23.4	25.4	27.0	27.9	28.2	24.4	21.4	17.9
Mean Daily Temperature Range	5% DB	WB	13.9	13.8	14.8	15.4	16.9	18.7	20.7	20.8	21.2	18.3	15.4	14.0
		MCDB	17.8	18.5	19.6	20.7	21.9	23.6	25.8	26.3	26.0	23.2	20.2	16.9
	5% WB	WB	13.0	13.0	14.0	14.6	16.1	17.9	20.0	20.3	20.3	17.5	14.8	13.1
		MCDB	16.9	17.5	18.1	19.3	20.5	22.3	24.5	25.2	24.6	21.8	19.2	16.2
Clear-Sky Solar Irradiance	taub	0.310	0.321	0.332	0.344	0.363	0.365	0.394	0.396	0.359	0.365	0.320	0.322	
	taud	2.556	2.539	2.491	2.462	2.428	2.424	2.429	2.402	2.483	2.437	2.554	2.549	
	Ebn at Noon	917	939	948	945	923	915	888	883	907	874	897	881	
	Edh at Noon	82	92	104	111	116	116	115	117	103	100	81	79	
All-Sky Solar Radiation	RadAvg	3.35	4.21	5.53	6.67	7.05	7.37	7.09	6.81	5.93	4.66	3.63	3.04	
	RadStd	0.21	0.35	0.30	0.34	0.51	0.42	0.46	0.31	0.32	0.33	0.23	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.85	N/A	N/A	+1.44	+0.76	+0.70	N/A	-190	+314	+140
(47)	Regional (33 neighbors)	+0.66	N/A	-1.61	+0.77	N/A	N/A	N/A	-106	+233	+126

Nomenclature: See separate page